Ammar Adel

Mechatronics & Robotics Researcher
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Education

Egypt-Japan University of Science and Technology (E-JUST) — Alexandria, Egypt
Bachelor of Science in Mechatronics and Robotics Engineering; CGPA: 3.8/4.0 (Excellent) — *Sep. 2021 – March 2026*

- **Relevant Coursework:** Automatic Control, Robotics, Selected Topics in Robotics, Artificial Intelligence in Robotics, Mobile Robots, Mechatronics System Design, Mechanical Vibrations, Mechanical Design, Pneumatics & Hydraulics, Embedded Systems, Manufacturing Processes, MEMS, Project Management.

Research & Publications

Undergraduate Researcher | Robotics Lab — Sep. 2024 – June 2026
Egypt-Japan University of Science and Technology (E-JUST) — *Alexandria, Egypt*

- **Advisor:** Prof. Ayman A. Nada
- **Fast-Tracking Gimbal System:** Developed kinematic models and advanced mechanical CAD designs for a 3-DOF real-time gimbal system engineered to track high-speed targets.
- **Stewart Platform Integration:** Integrated the gimbal system with a Stewart platform to simulate unstable dynamic environments and validate vision-IMU velocity control frameworks.
- **Publication:** Ammar Abdelwahab *et al.*, "Fuzzy Logic Control for Fast-Tracking Gimbals in Velocity Space," accepted for publication, *IEEE International Conference on Control, Decision and Information Technologies (CoDIT)*, 2026.

Control Systems Researcher — Feb. 2024 – Present
Egypt-Japan University of Science and Technology (E-JUST) — *Alexandria, Egypt*

- **Advisor:** Prof. Ayman A. Nada
- **Self-Balancing Robot Automation:** Scaled an Automatic Control course project into an advanced research initiative focused on the stabilization and dynamic control of a two-wheeled balancing robot.
- **Embedded Control Design:** Engineered and deployed real-time closed-loop control algorithms using NI myRIO, LabVIEW, and direct IMU sensor feedback.
- **Publication (In Preparation):** Conducting advanced system optimization and experimental data collection in preparation for an upcoming academic journal submission.

Industry Experience

R&D Mechatronics Engineer — May 2025 – Present
Oz-Tech Startup — *New Cairo, Egypt*

- **Agricultural Robotics Development:** Led the mechanical design and end-to-end manufacturing of autonomous agricultural systems, specifically Transplanter and Weeding robots.
- **Advanced CAD & Drafting:** Engineered 3D CAD models and 2D manufacturing drawings using SolidWorks and AutoCAD to transition prototypes into scalable production.
- **Actuation & Manufacturing:** Designed, simulated, and integrated complex pneumatic and hydraulic systems; managed part fabrication and assembly workflows.
- **Cross-functional Integration:** Collaborated with software and electrical engineering teams to integrate motion control systems, sensors, and mechanical hardware.

AI & Deep Learning - Afretec Summer Training — July 2024
American University in Cairo — *Cairo, Egypt*

- **Deep Learning Architectures:** Mastered the theoretical foundations and practical deployment of advanced neural networks, including CNNs, RNNs, and BERT utilizing TensorFlow and PyTorch.
- **Model Optimization:** Applied ensemble learning methods, hyperparameter tuning, feature engineering, and robust data balancing strategies to maximize inference accuracy on imbalanced datasets.
- **Edge AI Integration:** Bridged artificial intelligence and embedded hardware by deploying trained ML models onto microcontrollers through hands-on, hardware-in-the-loop interactive workshops.

Projects

Graduation Project: Design and Control of a Fast-Tracking Gimbal for Unstable Environments| [YouTube]

- Engineered and fabricated a 3-DOF fast-tracking gimbal system, optimizing the mechanical structure and kinematics for dynamic stability in unstable environments.
- Developed real-time closed-loop control algorithms (Fuzzy PID) and integrated IMU/vision sensor fusion to achieve high-speed, accurate target tracking.
- Validated the control frameworks and hardware reliability by simulating external disturbances through integration with a Stewart platform.

Mobile Manipulator (Autonomous Pick-and-Place) | *ROS 2, YOLOv5, Gazebo, Inverse Kinematics* | [GitHub]

- Engineered a 5-DOF robotic arm mounted on a 4-swerve-wheel autonomous mobile platform, utilizing SolidWorks for mechanical design and rapid prototyping.
- Implemented YOLOv5 for real-time object detection and integrated ROS 2 with Gazebo for kinematic simulation and autonomous path planning.

Electric Vehicle Rally (EVER V) | *SolidWorks, MATLAB, ANSYS* | [Credential]

- Led the mechanical design and FEA simulation of a high-performance steering system using SolidWorks and ANSYS, optimizing vehicle dynamics and handling.
- Led design and development of high-performance steering system
- Managed component procurement, systems integration, and funding, represented E-JUST among 27 shortlisted teams, proceeding to the competition with 21 finalist teams.

Self-Balancing Robot Automation | *NI myRIO, LabVIEW, PID/Fuzzy Control* | [YouTube]

- Modeled inverted pendulum dynamics and engineered the mechanical chassis to optimize the physical mass distribution for a two-wheeled balancing robot.
- Deployed real-time closed-loop control systems using NI myRIO and LabVIEW, tuning PID and Fuzzy Logic controllers with direct IMU feedback for dynamic stabilization.

MATE ROV Competition | *SolidWorks, Pneumatics, FluidSIM* | [Credential]

- Designed the hydrodynamic hull configurations and developed fully functional pneumatic grippers for a Remotely Operated Underwater Vehicle (ROV).
- Orchestrated the manufacturing process and integration of underwater sensors and control logic, ensuring rigorous adherence to competition specifications.

Multi-Legged Spider Robot | *Kinematics, Microcontrollers, SolidWorks* | [GitHub]

- Designed the mechanical linkages and structural prototype in CAD to achieve stable, biomimetic multi-legged locomotion.
- Synchronized multi-joint actuation by integrating embedded controllers, servo motors, and custom gait algorithms for effective terrain navigation.

Deep Q-Network (DQN) for Manipulator Control | *TensorFlow, Keras, Reinforcement Learning* | [GitHub]

- Trained a Deep Q-Network (DQN) algorithm to autonomously control a two-link robotic manipulator, guiding its joint angles to reach specific X, Y coordinate goals.
- Engineered a custom simulation environment to discretize action spaces and iteratively evaluated the model to minimize the number of steps required to reach the target.

EKF Mobile Robot Localization | *Python, Sensor Fusion, Mobile Robotics* | [GitHub]

- Programmed an Extended Kalman Filter (EKF) algorithm from scratch in Python to accurately localize a differential drive mobile robot within a mapped environment.
- Fused simulated sensor telemetry (distance and bearing of landmarks) with odometry to continuously estimate the robot's state and track its trajectory against ground truth.

5-DOF Robotic Arm Trajectory Control | *MATLAB, Simulink, Simscape Multibody, PID* | [GitHub]

- Modeled the rigid-body dynamics and joint mechanics of a 5-DOF articulated robotic arm utilizing MATLAB Simscape Multibody.
- Designed and tuned five independent continuous-time PID controllers in Simulink to execute precise joint-space trajectory tracking, successfully commanding the end-effector to draw a continuous circle.

Fast Object Tracking System | *LabVIEW, PD, Arduino, Servo Motor Control* | [YouTube]

- Designed an automated, closed-loop object tracking system utilizing a 5-channel IR sensor array and an Arduino Uno to dynamically direct a servo-actuated laser pointer at moving targets.
- Developed a real-time data acquisition and control interface in LabVIEW to process analog sensor feedback, calculate positional error, and compute immediate actuation corrections.

Honors & Awards

Dean's List & Academic Excellence Honor (CGPA: 3.8/4.0) | *E-JUST* March 2026

- Consistently recognized for elite academic performance in the Mechatronics and Robotics Engineering program, earning an "Excellent" distinction and A+ grades in all capstone graduation projects.

Undergraduate Research Grant ($1,500) | *E-JUST* Jan. 2026

- Secured highly competitive funding $1,500 to finance the hardware procurement, fabrication, and research development of the Fast-Tracking Gimbal graduation project.

2nd Place Winner – Go Dive Derby (ROV Competition) | *British University in Egypt (BUE)* July 2024

- Secured 2nd place and a 10,000 EGP prize against top national engineering teams by developing a highly capable Remotely Operated Vehicle (ROV); earned a qualification to represent Egypt at the global competition in the US.

Sawiris Distinction Scholarship (SDS) – Full Ride | *Amideast & Sawiris Foundation* Sep. 2021

- Awarded a highly competitive, fully funded scholarship to pursue a Bachelor of Engineering at E-JUST, selected based on exceptional academic merit, Thanaweya Amma ranking, and leadership potential.

Top 10 Ranked Student (Fayoum Governorate) | *Thanaweya Amma* July 2021

- Achieved a top 10 academic ranking across the entire governorate out of tens of thousands of students in the Egyptian General Secondary Education Certificate, demonstrating exceptional foundational knowledge in STEM.

Courses & Certificates

- Earned the **CB3 e-Learning Certificate** [Credential]
  from *Universal Robots Academy*, mastering the operation, programming, and safety configurations of robot arms.
- Completed an intensive 360-hour **Advanced Level English Course** [Credential]
  at *Amideast* to heavily refine technical communication and professional English proficiency.
- Selected as an **Afretec Summer Training Scholar** [Credential]
  at the *American University in Cairo (AUC)*, mastering deep learning and bridging AI models with embedded hardware.
- Achieved an **Embedded Systems Diploma** [Credential]
  through *Smart Technology*, developing expertise in Embedded C, ATmega32 microcontroller interfacing, and RTOS.

Technical Skills

- **CAD, Design:** SolidWorks, CATIA, AutoCAD, ANSYS, Adams, DFMA, Pneumatics, Sheet Metal, 3D Printing
- **Programming:** Python, C, ROS 2, PyTorch, TensorFlow, AI/ML/RL, Computer Vision (OpenCV, YOLO), Git, Linux
- **Simulation & Electronics:** MATLAB/Simulink/Simscape, LabVIEW, FluidSIM, Automation Studio, Proteus
- **Embedded & Control Systems:** NI myRIO, Microcontrollers (Raspberry Pi, ESP32, AVR), Control (PID, Fuzzy)
- **Languages:** Arabic (Native), English (Professional), German (Beginner), Japanese (Beginner)

Extra-Curricular Activities

Head of Steering Systems July 2022 – June 2025
E-JUST Racing Team *Alexandria, Egypt*

- **Shell Eco-marathon:** Designed and developed the steering architecture for an urban electric vehicle, successfully advancing through all technical inspection and competition stages.
- **Electric Vehicle Rally (EVER V):** Directed the end-to-end design and manufacturing of the steering system for E-JUST's inaugural EV rally participation.
- **Engineering Application:** Applied SolidWorks, MATLAB, ANSYS, and Adams to conduct rigorous 3D modeling, FEA simulation, and vehicle dynamics analysis.

Technical Member and Instructor March 2023 – Feb. 2025
E-JUST Robotics Club *Alexandria, Egypt*

- **Workshops:** Mentored 30+ students through comprehensive robotics workshops, covering entry-level to advanced mechatronics concepts.
- **Projects:** Contributed to the Aimbot and Smart Chess Board projects, successfully integrating hardware, software, and intelligent computer vision with traditional gameplay.
- **Competitions:** Competed in the MATE ROV and OpenCV competitions, specializing in underwater robotics mechanics and computer vision algorithms.